Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 100308 | *Subject: Electromagnetic Fields* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) The dot product of two vectors is zero when:
- (i) They are parallel
- (ii) One is a scalar
- (iii) They are perpendicular
- (iv) Their magnitudes are equal

(b) Coulomb's law describes the force between:
- (i) Two magnetic poles
- (ii) Two stationary point charges
- (iii) Two current-carrying wires
- (iv) A moving charge and a magnetic field

(c) Current density $\vec{J}$ is related to current $\vec{I}$ by:
- (i) $I = \vec{J}.\vec{A}$
- (ii) $\vec{J} = I^2\vec{A}$
- (iii) $I = \vec{E}.\vec{A}$
- (iv) $\vec{J} = \frac{I}{A^2}$

(d) The unit of magnetic flux is:
- (i) Tesla
- (ii) Farad
- (iii) Henry
- (iv) Weber

(e) The force on a moving charge in a magnetic field is given by:
- (i) $\vec{F} = q\vec{E}$
- (ii) $\vec{F} = q\vec{B}$
- (iii) $\vec{F} = q\vec{v} \times \vec{B}$
- (iv) $\vec{F} = \vec{v} \times \vec{E}$

(f) Faraday's law of electromagnetic induction states that:
- (i) Magnetic field induces electric current
- (ii) Changing electric field induces magnetic field
- (iii) Changing magnetic flux induces electromotive force
- (iv) Magnetic field is proportional to current

(g) The wave equation in free space is derived from:
- (i) Ohm's law
- (ii) Faraday's law only
- (iii) Biot–Savart Law
- (iv) Maxwell's equations

(h) The cross product of two vectors is a:
- (i) Vector
- (ii) Scalar
- (iii) Tensor
- (iv) Matrix

(i) The relative permittivity ε_r of a material affects:
- (i) Current in a conductor
- (ii) Capacitance of a capacitor
- (iii) Magnetic field strength
- (iv) Resistance of a wire

(j) The displacement current was introduced by Maxwell to:
- (i) Account for magnetic field in conductors
- (ii) Explain magnetic flux in insulators
- (iii) Ensure continuity of current in a capacitor
- (iv) Explain dielectric heating

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Q.2	(a)	Describe the procedure for converting a vector from one coordinate system from cylindrical to rectangular coordinates. Show all necessary steps with an example.	[7]
	(b)	State and explain Gauss's Divergence Theorem and Stokes' Theorem.	[7]
Q.3	(a)	Define electric field intensity. Derive an expression for the electric field due to a point charge?	[7]
	(b)	Derive the expression for the potential difference in the following configurations: (i) Between two point charges (ii) Due to a uniformly charged ring on its axis	[7]
Q.4	(a)	Define capacitance. Derive an expression for the capacitance of a parallel plate capacitor with dielectric.	[7]
	(b)	State and derive Laplace's and Poisson's equations from Gauss's law. Under what conditions does each apply?	[7]
Q.5	(a)	Define magnetic flux and magnetic flux density. Derive the relation between $\vec{B}$ and $\vec{H}$?	[7]
	(b)	Derive the expression for the magnetic field produced by a finite length straight conductor using Biot-Savart Law.	[7]
Q.6	(a)	Derive the force experienced by a differential current element placed in a magnetic field.	[7]
	(b)	Describe the different types of magnetic materials (diamagnetic, paramagnetic, and ferromagnetic) and their behavior in an external magnetic field.	[7]
Q.7	(a)	Write all four Maxwell's equations in both point form and integral form. Briefly describe the physical meaning of each.	[7]
	(b)	Derive the expression for motional EMF when a conductor moves in a magnetic field.	[7]
Q.8	(a)	State and prove the Poynting theorem.	[7]
	(b)	What is the skin effect? Derive the expression for skin depth and explain its dependence on frequency and material properties.	[7]
Q.9	(a)	Define current and current density. Derive the relationship between them and explain how current density varies in a non-uniform conductor.	[7]
	(b)	Define magnetization and magnetic susceptibility. Derive the relation between $\vec{B}$, $\vec{H}$, and $\vec{M}$.	[7]

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